

RESEARCH SUMMARY

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In Section 1, I discuss projects whose writeups have drafts and are close to completion. I also have some ideas that I am very excited about that are in various stages of completion, which I discuss more in Section 2. I hope to make some headway on these in the coming months.

1. PAST/PRESENT

My graduate work to date broadly falls into two areas:

- Computing Hilbert modular forms and Hilbert modular varieties.
- Understanding the relationships between atypical (non-generic Mumford–Tate group) abelian fourfolds and automorphic forms.

These are not unrelated lines of work. The Hilbert modular forms I am most interested in computing – non-CM forms of weights $(1, 2)$ or $(1, 1, 2)$ (see Section 2) – are among the automorphic forms I am trying to associate to certain classes of atypical abelian fourfolds through (potential) automorphy theorems. In some sense, the origin of both lines of work was a [blog post](#) of Frank’s, where he posits a relationship between non-CM weight $(1, 1, 2)$ Hilbert modular forms and abelian fourfolds of “Mumford’s type”. In what follows, I briefly describe each of my projects and provide links.

Though this may change at some point going forward, I have not used AI seriously in my work so far beyond literature search, proofreading papers, and (in the last few weeks) writing code.

1.1. *Computing Hilbert modular forms with nonparititious weight*, to appear in the Proceedings of ANTS XVII.

Recall that a Hilbert modular form (HMF) over a degree n field has parititious weight if its weight is $(k_1, \dots, k_n)$ with $k_1 \equiv k_2 \equiv \dots \equiv k_n \pmod{2}$. Historically, theory and computation have focused on the setting of parititious weight, as in this case the associated automorphic representation is algebraic. We devise and implement an algorithm which, given a field F with narrow class number 1, a nonparititious weight k , a level $\mathfrak{N} \subset \mathbb{Z}_F$, and a nebentypus character χ of modulus $\mathfrak{N}$, computes a basis of q -expansions spanning the space of cusp forms $S_k(\mathfrak{N}, \chi)$. **The main obstacle is that the usual Hecke operators $T_{\mathfrak{p}}$, for $\mathfrak{p} = (\pi)$ a prime ideal in $\mathbb{Z}_F$, can only be defined over a field containing a square root of π . In particular, the usual Hecke algebra $\mathbb{T}$ is not a finite algebra over a field K for any finite extension $K/\mathbb{Q}$.** Instead, we introduce and work with the “elemental Hecke algebra” $\mathbb{T}^{\text{elem}}$, which is a twist of the usual Hecke algebra that *is* defined over a number field and agrees with the usual Hecke algebra when k is parititious. In studying the elemental Hecke algebra, we also give self-contained proofs of many standard results in HMF theory.

1.2. *Computing Hilbert modular forms with nontrivial nebentypus*, joint with Eran Assaf.

One can compute a basis for a space of HMFs by computing matrices for the Hecke action on the space and using these matrices to produce a basis. There are two main methods for this – the definite method, which generalizes the classical method of Brandt matrices, and the indefinite method, which computes the Hecke action on group cohomology using the Eichler–Shimura isomorphism. This paper extends both of these methods to Hilbert modular forms with nontrivial nebentypus and provides implementations. The theory in this paper is largely “routine to the experts”, extending the machinery of Dembélé–Voight [DV11] to these two cases. However, beyond the technical content and the implementation, I think this paper has expository value. First, it is a useful primer on studying automorphic forms on $B^\times$. Secondly, it provides a short self-contained proof of the Eichler–Shimura isomorphism that (while implicit in Matsushima–Shimura [MS63], who prove a much more general result) I have not seen in this form elsewhere.

1.3. *Computing with Fourier expansions of Hilbert modular forms*, joint with many co-authors.

(Editing, hopefully will be arXiv-ready soon!) This is a longer paper which does several things. We develop efficient algorithms for manipulating (and in particular, multiplying) q -expansions of HMFs. We first use efficient multiplication to produce an algorithm for computing presentations of graded rings of HMFs – these are (roughly) rings $S(\mathfrak{N}) := \bigoplus_{k \in 2\mathbb{Z}_{\geq 0}} M_k(\Gamma_0(\mathfrak{N}))$, and $\text{Proj } S(\mathfrak{N})$ is a model for the Baily–Borel compactification of $X_0(\mathfrak{N})$. Numerous previous papers (e.g. [Gun63, Hir77, vdGZ77, EK14]) have computed presentations for individual graded rings or models for individual varieties, but **our algorithm gives a way to compute these objects “at the press of a button”**. The rings $S(\mathfrak{N})$ are finitely presented, and we can compute a basis for arbitrarily many graded pieces with existing technology, so **the key challenge is knowing when we have enough generators and can stop**. Using the trace formula, we can compute the Hilbert series of $S := S(\mathfrak{N})$. However, even if a candidate graded ring S_D (produced from the generators and relations in degree at most D) has the same Hilbert series as S , we could still be missing generators and relations in higher degrees. **The idea is to use the explicit q -expansions to compute the Jacobian (analytically) of an appropriate map to $\mathbb{C}^n$ (where n the dimension of the Hilbert modular variety) and use it to show that the image of S_D in S has the appropriate dimension**. We also implement the Hecke stability method for Hilbert modular forms, and **use it to produce the first example of a parallel weight $(1, 1)$ Hilbert modular form with non-dihedral projective image**.

1.4. *Galois representations of abelian varieties with nontrivial geometric endomorphisms*.

(Editing, hopefully will be arXiv-ready very soon!). The first part of this paper gives a structure theorem for the compatible system $\mathcal{V}$ of Galois representations associated to a g -dimensional geometrically simple abelian variety A/F with nontrivial geometric endomorphism algebra D . Let r be the index of D and e the degree of the center E of D , and let K/F be the minimal field over which $E \hookrightarrow \text{End}_K^0(A)$. Then, there is a compatible system $\mathcal{R}$ of $\frac{2g}{re}$ -dimensional strongly irreducible G_K -representations and a compatible system $\mathcal{A}$ of r -dimensional Artin G_K -representations, such that $\mathcal{R}$ and $\mathcal{A}$ completely determine $\mathcal{V}$. Letting L/F be the field over which all the geometric endomorphisms of A are defined, it is not at all surprising that there are compatible systems of $\frac{2g}{re}$ -dimensional strongly irreducible G_L -representations determining $\text{Res}_{G_L} \mathcal{V}$ – writing $\rho_{A,\ell}$ for the ℓ -adic representation associated to A , there exist $\{\xi_i\}_{1 \leq i \leq e}$ such that $\text{Res}_{G_L} \rho_{A,\ell} \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{1 \leq i \leq [E:\mathbb{Q}]} \xi_i^{\oplus r}$ because $D \otimes \overline{\mathbb{Q}_\ell} \cong M_r(\overline{\mathbb{Q}_\ell})^e$. **The point is that this compatible system $\mathcal{R}$ is a compatible system of G_K -representations and not just G_L -representations. In many interesting cases, $K = F$ even though L/F may be much larger**. This structure theorem lets us reprove many results about the fields of definition of endomorphisms of abelian varieties. Previous work of Fité–Florit–Guitart [FFG22] gives a structure theorem for $\text{Res}_{G_{K'}} \mathcal{V}$, where K'/K is the smallest field where $\text{Res}_{G_{K'}} \mathcal{A}$ has cyclic projective image. In the second part of this paper, we use the structure theorem to prove potential modularity theorems for several families of abelian fourfolds over totally real fields with nontrivial geometric endomorphisms. Let F be totally real. **In the most interesting cases, $\mathcal{R}$ is four-dimensional and $K = F$ or K is real quadratic, and we apply the modularity lifting theorems of Boxer–Calegari–Gee–Pilloni [BCGP18] with some modifications to the Moret–Bailly arguments in order to prove potential modularity**. In the first of these cases, when D is an indefinite quaternion algebra, we cannot use the usual PEL type Shimura variety Y as a moduli space because the endomorphisms of A may not be defined over a totally real field (and hence A cannot be used to prove non-emptiness of $Y(\mathbb{R})$). **Instead we apply Moret–Bailly to a twist Y' of the usual PEL moduli variety produced via Galois descent, on which A is a real point**. When D is a real quadratic field, finding enough ordinary primes becomes an issue. **It is crucial that $K = F$ is totally real here because it allows us to apply potential modularity theorems for representations over totally real fields, even though K' and L may not be totally real**. Lastly, we apply the structure theorem and a construction of Frank Calegari to show that there exist geometric four dimensional symplectic representations with big image which are even. Calegari [Cal11] showed that all such two-dimensional representations are odd, so this is (to me) quite surprising.

2. NEAR FUTURE

Here are projects that I hope to work more on in the coming months. In all cases, these are projects that I believe I have made nontrivial progress on, but where there might still be some math to work out or some uncertainty. I hope to at least have a draft of the first project ready soon.

2.1. Potential modularity of abelian fourfolds with geometric endomorphisms by a definite quaternion algebra.

For simplicity, let $A/\mathbb{Q}$ be an abelian fourfold with geometric endomorphisms by a definite quaternion algebra D with center $\mathbb{Q}$. Applying the structure theorem of the previous paper, $\mathcal{V} \cong \mathcal{R} \otimes \mathcal{A}$ where $\mathcal{R}$ is a compatible system of 4-dimensional $G_{\mathbb{Q}}$ -representations preserving orthogonal pairings up to similitude. Therefore, $\rho_{A,\ell} \cong \xi \otimes \varrho$ for a 4-dimensional representation $\xi: G_{\mathbb{Q}} \rightarrow \mathrm{GO}_4(\overline{\mathbb{Q}}_{\ell})$ and a 2-dimensional Artin representation ϱ . **I claim that this ξ is (up to twist) the Galois representation associated to $\mathrm{Asai}(\pi_f)$, where π_f is the automorphic representation associated to a Hilbert modular form f over a real quadratic field F of weight $(1, 2)$.** Note that because f has nonparitious weight, π_f is not algebraic and should not have an associated compatible family of geometric Galois representations [BG14]. However, $\mathrm{Asai} \pi_f$ is algebraic, and has the appropriate Hodge–Tate weights. While π_f is not algebraic as a GL_2 -representation, it is algebraic as an SL_2 -representation, and we can associate to it a projective Galois representation [Pat19]. **The main difficulty is proving (potential) automorphy of 2-dimensional projective Galois representations that look like they should come from weight $(1, 2)$ Hilbert modular forms.** For this, we use Hida theory, in the style of Buzzard–Taylor’s proof of modularity for weight 1 modular forms [BT99]. However, because the relevant Hilbert modular forms are nonparitious and partial weight 1, there are many technical difficulties that still need to be addressed.

2.2. Computing non-CM weight $(1, 2)$ and $(1, 1, 2)$ Hilbert modular forms.

This was the initial motivation behind many of my current projects. I have code that computes $(1, 2)$ and $(1, 1, 2)$ spaces, but have been unable to actually find a non-CM example so far. I hope to optimize the indefinite method further using the quasilinear time algorithm of Bařt and Page [BP26] for computing generators of arithmetic Fuchsian groups, and use more computational resources to extend my searches.

2.3. Restriction to the diagonal of dihedral weight $(1, 1)$ Hilbert modular forms.

I think that every weight 2 modular newform g with rational coefficients and analytic rank 0 or 1 arises as the restriction to the diagonal of a weight $(1, 1)$ Hilbert modular form f with dihedral projective image induced from a CM extension K/F which is biquadratic over $\mathbb{Q}$. Without too much computational effort, I’ve been able to verify this for all elliptic curves with conductor at most 50 and 341 of the 475 isogeny classes of elliptic curves with conductor at most 300. Applying a theorem of Fornea [For19], one can show that the Petersson inner product $\langle \mathrm{Res} f, g \rangle$ is (up to scalars) equal to

$$L(1, g \times \mathrm{Asai} f) = L(1, g \times \mathrm{Ind}_{G_{K_1}}^{G_{\mathbb{Q}}} \chi_1) L(1, g \times \mathrm{Ind}_{G_{K_2}}^{G_{\mathbb{Q}}} \chi_2) = L(1, (\mathrm{Res}_{G_{K_1}} \rho_g) \otimes \chi_1) L(1, (\mathrm{Res}_{G_{K_2}} \rho_g) \otimes \chi_2),$$

for imaginary quadratic fields $K_i/\mathbb{Q}$ and finite-order characters χ_i of G_{K_i} . These twisted L -values have been related to Heegner points, e.g. in work of Michel–Venkatesh [MV05], though I’m not sure if there is a connection here. Even setting aside generality, **the fact that the L -function of even one non-CM elliptic curve (which I think of as a “random” object) can be obtained by starting with a very “structured” object (a finite-order character of a CM biquadratic extension of $\mathbb{Q}$) and applying a “noising” operation (restriction to the diagonal of the induced HMF, which aggregates terms with the same trace) is very striking to me.**

2.4. Modularity of an example of a weight 2 motive with $h^{2,0} = 2, h^{1,1} = 0$ with Hodge Lie algebra $\mathfrak{sl}_2^{\oplus 2}$, joint with Frank Calegari.

Griffiths transversality predicts that weight 2 motives with $h^{2,0} = 2, h^{1,1} = 0$ should not vary in families, so it is interesting to produce examples of them. While CM examples can be produced by considering appropriately branched covers of the projective plane and examples with Hodge Lie algebra $\mathfrak{sl}_2$ arise from appropriate modular curves $X_1(N)$, an example with $\mathfrak{sl}_2^{\oplus 2}$ was not known to exist. Recently, ChatGPT 5.6, prompted by Frank Calegari and building on ideas of David Zywinia, produced a construction of an elliptic surface whose transcendental lattice is exactly such a motive. (This construction has not yet been verified by a human.) Some years ago, I computed a weight $(1, 3)$ Hilbert modular form f over $F = \mathbb{Q}(\sqrt{5})$ with L -polynomials satisfying rationality properties suggesting that the tensor induction $\bigotimes_{G_F}^{G_{\mathbb{Q}}} \rho_f$ should be associated to exactly such a motive, and it is not hard to search for more. **Since we now have a way to compute $(1, 3)$ forms and a strategy for constructing these elliptic surfaces, it should be possible to find an elliptic surface with 4-dimensional Galois representation ρ (on the transcendental**

lattice) with Hodge–Tate weights $(0, 0, 2, 2)$ and a weight $(1, 3)$ Hilbert modular form f such that $\rho \cong \bigotimes_{G_F}^{G_{\mathbb{Q}}} \rho_f$.

2.5. (More) explicit examples of Shimura curves parameterizing geometrically type III abelian fourfolds.

The Shimura curves associated to abelian fourfolds of so-called “Mumford’s type” [Mum69] are associated to a quaternion algebra B/F , where F is a totally real cubic field and B is ramified at exactly two of the three infinite places. Noot [Noo01] then classified the CM points on these curves. I can show that **the PEL Shimura variety associated to abelian fourfolds with endomorphism algebra containing a definite quaternion algebra $D/\mathbb{Q}$ arises via an analogous construction from a quaternion algebra B/F , where F is real quadratic, B is ramified at exactly one place, and $\text{Cor}_{F/\mathbb{Q}} B$ is Brauer equivalent to D .** I expect an argument completely analogous to Noot’s to classify the CM points on these curves – it should also not be hard to recover analogues of his other results for Mumford-type Shimura curves in the setting of these Shimura curves. Explicit examples of Mumford type abelian fourfolds were computed by Bouchet–Hanselman–Pieper–Schiavone [BHPS25] by first using Noot’s classification to find CM points and then interpolating using invariants for genus 4 curves to produce a model for the Shimura curve. I hope to do the same in this setting. In this case, examples of such Shimura curves have already been constructed by Cantoral-Farfán–Lombardo–Voight [CFLV23], but it would be interesting to produce these curves in different ways.

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